

Evaluating Trading Real Optionality Between NGLs and Petrochemicals

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Abstract

Energy commodity trading strategies often exploit the real optionality embedded in physical transformation processes, such as refining one energy source into another. In the natural gas liquids (NGLs) sector, the ability to convert NGLs into petrochemicals creates opportunities for statistical arbitrage strategies that systematically capture and capitalize on price dislocations. This study examines whether systematically monitoring processing spreads between an NGL and its petrochemical derivative can lead to unique trading strategies. By evaluating both intra- and inter-spread strategies, the analysis identifies the most effective approaches and highlights the structural market dynamics that drive persistent and stable returns.

1. Introduction

Natural gas liquids (NGLs) are a class of hydrocarbons found in natural gas and crude oil reservoirs. Initially in a gaseous state, these compounds condense into liquids under specific temperature and pressure conditions and serve as both energy sources and key feedstocks for petrochemical production. NGLs are separated from natural gas and crude oil during midstream processing through fractionation, which isolates hydrocarbons based on their boiling points. The primary NGLs traded as futures include ethane (C_2H_6), propane (C_3H_8), butane (C_4H_{10}), isobutane (C_4H_{10} , branched), and a broader mix of heavier hydrocarbons ($C_5H_{12}^+$). In petrochemical production, steam cracking subjects ethane, propane, and butane to high temperatures, breaking them down into high-value petrochemicals such as ethylene (C_2H_4) and propylene (C_3H_6), essential building blocks for plastics and chemicals. Additionally, propane dehydrogenation (PDH) serves as an alternative pathway for propylene production.

These processing relationships create a real optionality framework within the NGL value chain, where shifting market conditions dictate the most economically viable production pathway. The interplay between fractionation, steam cracking, and dehydrogenation links NGL and petrochemical markets, allowing for arbitrage opportunities based on feedstock substitution and margin fluctuations. These interconnections between futures markets support “quantamental” trading strategies which integrate systematic execution with discretionary economic insights to identify and capitalize on structural price relationships.

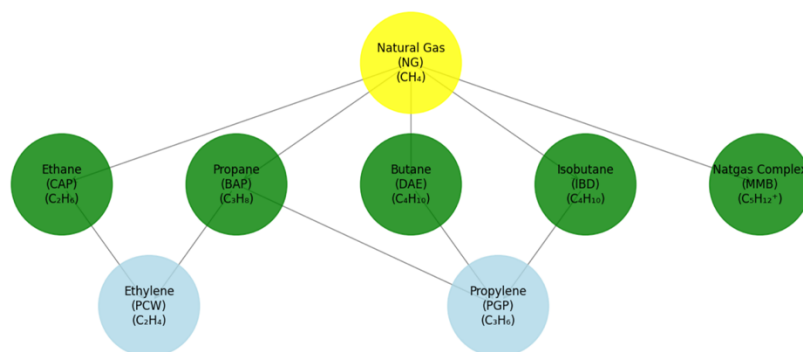


Figure 1: Real Optionality Structure

To focus on trading within the niche NGL-specific market, spreads are constructed along the real option refining links between NGLs and their derived petrochemicals. Spreads involving upstream natural gas are excluded from strategy construction due to a geographic disconnect—natural gas futures are priced at Henry Hub, whereas NGLs and petrochemicals are priced at Mont Belvieu. Moreover, incorporating natural gas spreads would require adjusting each NGL for its equivalent MMBTU conversion, introducing additional uncertainty. The selected spreads, therefore, focus on localized optionality within Mont Belvieu, where steam cracking and dehydrogenation economics are more directly observable. The selected spreads are:

- **Ethane (CAP) – Ethylene (PCW):** Ethane is the most efficient and cost-effective feedstock for ethylene.
- **Propane (BAP) – Ethylene (PCW):** Propane becomes a viable ethylene feedstock when ethane is scarce or overpriced. This spread tracks shifts in cracker feedstock preferences and propane's discount to ethane.
- **Propane (BAP) – Propylene (PGP):** Dehydrogenation (PDH) converts propane into propylene, making this spread a key indicator of PDH profitability.
- **Butane (DAE) – Ethylene (PCW):** Butane can be cracked into ethylene, but its higher carbon content makes it less efficient.
- **Butane (DAE) – Propylene (PGP):** Butane can also produce propylene via catalytic cracking.

2. Data

This study utilizes a dataset comprising daily swaps, swap futures, and futures contract prices for the commodities within the presented real optionality structure, covering the first 24 monthly contracts (F1 to F24) to allow for wider futures term structure analysis over time. These instruments are treated as comparable forward-looking price indicators for the same delivery periods, and the term "futures" will be used interchangeably to refer to any of these three. However, differences in liquidity, counterparty risk, and settlement mechanisms should be noted, as swaps and swap futures may exhibit slight deviations from exchange-traded futures due to their bilateral nature and margining structures.

Commodity	Contract Type	Ticker	Location	Units
Ethane	Generic 1st Ethane, Mt Bel 5 Dec Swap (NYM)	CAP	Mont Belvieu	(¢/gal)
Propane	Generic 1st Propane, Mt Bel 5 Dec Swap (NYM)	BAP	Mont Belvieu	(¢/gal)
Butane	Generic 1st Butane, Mt Bel 5 Dec Swap (NYM)	DAE	Mont Belvieu	(¢/gal)
Isobutane	Generic 1st Isobutane, Mt Bel 5 Dec OPIS Swap Future (NYM)	IBD	Mont Belvieu	(¢/gal)
Ethylene	Generic 1st Mt Be Ethylene Futures (NYM)	PCW	Mont Belvieu	(¢/lb)
Propylene	Generic 1st Polymer Grade Propylene (PCW) Futures (NYM)	PGP	Mont Belvieu	(¢/lb)

Table 2: Data Overview

All data is sourced from Bloomberg. The sample period begins on July 7, 2011, aligning with the earliest available first month (F1) trade date for Ethylene (PCW)—the last commodity within the analyzed group to undergo financialization. The dataset spans nearly 14 years, concluding on February 24, 2025, and comprises approximately 3,500 daily observations per time series.

Price quotes differ between commodity groups. NGLs are quoted in dollars per gallon (\$/gal), while petrochemicals are quoted in dollars per pound (\$/lbs). To ensure like-for-like comparisons in spread construction, petrochemical prices are converted to cents per gallon using:

$$\text{Price in \$/gal} = \text{Price in (\$/lbs)} \times \text{Density (lb/gal)}$$

The approximate densities used are¹:

- Ethylene ≈ 4.91 lb/gal
- Propylene ≈ 4.16 lb/gal

spread construction is performed manually, with spreads defined as the difference between the front-month (F1) futures of the petrochemical and the front-month futures of the corresponding NGL. Lagging the NGL to account for processing time is not required, as it is assumed that refineries hold existing stock within their storage facility. The general crack spread formula is:

$$\text{Processing Spread} = F1_{\text{Petrochemical},(t)} - F1_{\text{NGL},(t)}$$

To practically replicate holding any single underlying commodity, a rolling futures strategy is implemented at the end of each month to maintain continuous, practical exposure. This involves selling the first month (F1) futures contract and buying the second month (F2) contract, with a theoretical transaction cost of 0.2% of the F1 contract value at each sale. For example, a position held in January would involve selling the January contract on the 31st, purchasing the February contract for delivery on February 28th, and recognizing that on February 1st, the February contract naturally becomes the new F1. This method ensures continuous exposure while accounting for the costs associated with rolling futures contracts.

3. Regime Analysis for Strategy Construction

Understanding historical regime behavior is essential for constructing effective statistical arbitrage strategies between NGLs and petrochemicals. Before the shale boom (pre-2015), markets exhibited high volatility and persistently negative roll yields, driven by a normal backwardation structure. In this regime, producers were the dominant hedgers, selling futures contracts to manage price risk. To attract liquidity from speculators, these futures were priced at a discount to expected spot prices, creating a structural risk premium. As a result, traders holding long positions faced consistent roll yield losses as contracts approached expiration.

¹ Density conversion values are sourced from the industrial database *Engineering Toolbox*

The shale boom era (2015-2021) introduced greater market stability as increased production reduced supply constraints. Crack spreads tightened due to improved refining efficiencies and steady availability, while roll yields moved closer to breakeven as speculative participation increased. Spread deviations became less frequent, but when they did occur, they were more pronounced due to stronger mean-reverting tendencies.

The COVID-19 shock in 2021 triggered extreme price volatility, briefly disrupting the normal backwardation structure. Sharp price surges temporarily offset the negative carry associated with rolling futures, creating unique arbitrage opportunities. Crack spreads widened significantly but remained dislocated for only short periods before reverting. Since 2022, markets have stabilized to shale-era levels, with negative roll yields persisting at marginal but steady levels. Crack spreads have remained relatively tight, though propylene spreads continue to trade at a premium to ethylene, reflecting a structural shift first observed during the shale boom.

The profitability of rolling futures is closely tied to these regimes. Negative roll yields dominate during stable periods of normal backwardation, while market disruptions can briefly create profitable windows for long positions. Similarly, crack spreads exhibit regime-dependent behavior, with tighter spreads in stable periods and wider deviations during price shocks. This alignment of roll yield performance and spread behavior underlines the importance of regime awareness in trading strategies.

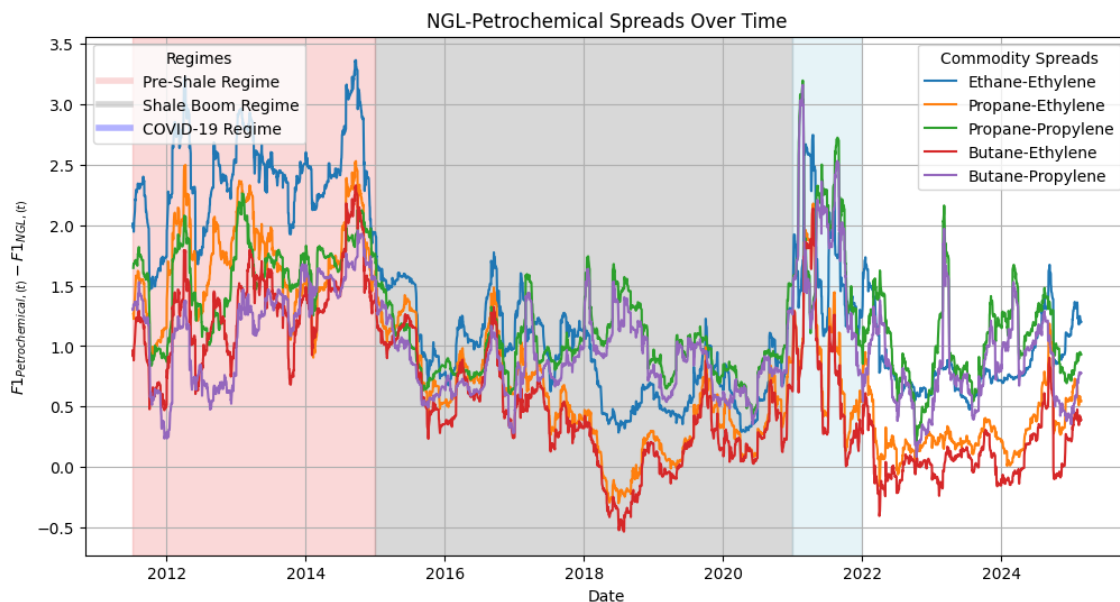


Figure 2: Spreads over time

Given this structure, mean-reversion strategies must be robust across shifting market conditions to remain effective. To ensure stability, Engle-Granger cointegration tests are applied at the 5% significance level across the entire historical period. Only pairs exhibiting persistent cointegration across multiple regimes are selected, filtering out transient relationships that could

lead to unreliable trading signals or premature exits during market transitions. By incorporating regime awareness into the statistical arbitrage framework, strategies can better withstand market shifts and focus on pairs with true long-term relationships, ensuring that trading decisions are grounded in fundamental market dynamics.

Spread	Statistic	P-Value	Spread Pair	Statistic	P-Value
Ethane-Ethylene	-2.945926	0.12508	Ethane-Ethylene vs Propane-Ethylene	-3.076768	0.09948
Propane-Ethylene	-2.792506	0.23113	Ethane-Ethylene vs Propane-Propylene	-3.876692	0.00274
Propane-Propylene	-5.027702	0.00079	Ethane-Ethylene vs Butane-Ethylene	-2.941496	0.10280
Butane-Ethylene	-3.361138	0.04685	Ethane-Ethylene vs Butane-Propylene	-2.873921	0.14588
Butane-Propylene	-4.628376	0.00018	Propane-Ethylene vs Propane-Propylene	-3.590144	0.01932
			Propane-Ethylene vs Butane-Ethylene	-3.453686	0.03273
			Propane-Ethylene vs Butane-Propylene	-2.756253	0.11782
			Propane-Propylene vs Butane-Ethylene	-5.306008	0.00007
			Propane-Propylene vs Butane-Propylene	-3.026519	0.10977
			Butane-Ethylene vs Butane-Propylene	-3.408403	0.03570

Table 3: Engle-Granger Cointegration Tests

The commodities and spreads for the following strategies will therefore be constrained to only those that show long-run cointegration over different regimes.

To avoid lookahead bias, the following trading indicators will be explicitly lagged to ensure that it relies only on past information. Since the dataset reflects the last price of each commodity, this lag is essential because traders cannot act on the same day's closing price data. Instead, they execute trades based on the information available from the prior trading day. This approach ensures that the strategy reflects a realistic trading environment and that all signals are based on observable and actionable data. Consequently, this structure prevents inadvertent overfitting or exploitation of information that would be inaccessible in a real-time market scenario, thereby preserving the integrity and validity of the strategy's back testing results.

3.1 Intra-Spread Mean Reversion

The intra-spread mean reversion statistical arbitrage strategy is based on the real optionality to switch between different NGL feedstocks for the same petrochemical output. Since ethylene and propylene can be produced from multiple feedstocks, refiners dynamically optimize their input choices based on market pricing and refining margins. As a result, the spread between an NGL and its corresponding petrochemical derivative should remain bounded over time, as economic incentives drive refiners to switch feedstocks when the cost differential becomes too large. This fundamental relationship is confirmed by cointegration tests, supporting the validity of a mean-reversion strategy that profits from deviations from equilibrium.

The trading signal for the intra-spread strategy is defined as:

$$\pi(S_t) = \begin{cases} -1 & \text{if } S_t - MA_t(S_t; n) > \varepsilon \\ +1 & \text{if } S_t - MA_t(S_t; n) < -\varepsilon \\ 0 & \text{otherwise} \end{cases}$$

where:

- (S_t) is the observed spread at time (t) ,
- $(MA_t(S_t; n))$ is the moving average of the spread over a look-back window (n) ,
- (ε) is the threshold to filter significant deviations.

The moving average serves as a proxy for equilibrium spread levels, reflecting shifts in refining infrastructure and long-term cost structures. The threshold (ε) acts as a buffer against short-term noise, ensuring that only economically meaningful mispricings trigger trades. Various look-back periods and threshold values are tested to optimize signal generation, with parameter selection guided by Sharpe ratio maximization.

From a practical perspective, this strategy simulates a structured feedstock switching option. Refiners continuously optimize feedstock selection based on cost efficiency. If one feedstock becomes too expensive relative to its output, refiners shift to lower-cost alternatives, increasing demand for the cheaper input while reducing demand for the expensive one. This adjustment compresses the spread, reinforcing mean-reverting behavior. To maintain continuous exposure, positions are rolled forward at expiration. By systematically exploiting deviations in feedstock pricing, this strategy mirrors the decision-making process of refiners and arbitrageurs who optimize between ethane, propane, and butane as feedstocks, reinforcing the economic intuition behind mean-reverting price behavior in intra-spreads.

3.2 Inter-Spread Mean Reversion

Beyond individual crack spreads, this strategy extends the analysis to inter-spread relationships, specifically the relative value dynamics between multiple crack spreads. This approach is based on the optionality to switch between producing different petrochemical outputs using different feedstocks, reflecting the flexibility in real-world refining operations. Petrochemical producers adjust both feedstock inputs and output mixes based on relative profitability. When one refining pathway becomes significantly more profitable than another, producers shift accordingly, which reduces the differential between spreads and reinforces mean-reverting behavior.

The trading signal for the inter-spread strategy is defined as:

$$\theta(R_t) = \begin{cases} -1 & \text{if } R_t - MA_t(R_t; n) > \varepsilon \\ +1 & \text{if } R_t - MA_t(R_t; n) < -\varepsilon \\ 0 & \text{otherwise} \end{cases}$$

where:

- (R_t) is the differential between two crack spreads at (t)

- $(MA_t(R_t; n))$ is the moving average of the spread differential over a look-back window (n)
- (ϵ) is a threshold to reduce sensitivity to minor noise as before

The trading signal for the inter-spread strategy follows a similar framework to the intra-spread strategy but is applied to the differential between two crack spreads. The moving average serves as a dynamic proxy for equilibrium, adapting to shifts in refining economics while filtering short-term noise. Since crack spreads adjust over time based on cost structures and arbitrage dynamics, a properly chosen moving average length balances capturing persistent trends with avoiding false signals. As refiners adjust both feedstock selection and petrochemical output in response to shifting market conditions, the relative difference between crack spreads should remain bounded over time. Deviations from this equilibrium present opportunities to take positions in anticipation of mean reversion.

Unlike the intra-spread strategy, which directly trades the price relationship between an NGL and its derivative, the inter-spread strategy takes a position in the spread differential itself. A positive signal indicates that one spread is undervalued relative to another, leading to a long position in the higher-margin spread and a short position in the lower-margin spread. Conversely, a negative signal suggests that the relationship has deviated in the opposite direction, inverting the position. This strategy involves maintaining rolling futures positions in both petrochemical and NGL legs, constructing a four-leg spread trade. The rolling mechanism ensures continuous exposure, with execution optimized to minimize slippage and roll yield costs.

For example, a positive signal on the Propane-Ethylene vs Propane-Propylene spread suggests that propane-ethylene margins are temporarily compressed relative to propane-propylene. To exploit this mispricing, the strategy goes long propane-ethylene (long ethylene, short propane) and short propane-propylene (short propylene, long propane), expecting convergence over time while maintaining continuous exposure. This structure ensures that the trade reflects the expectation of spread convergence while maintaining continuous exposure to price movements.

By capturing both feedstock substitution and output flexibility, the inter-spread strategy enables a more refined approach to arbitrage. Rather than relying on the mean-reverting tendencies of individual crack spreads, it identifies distortions in the relative pricing of alternative refining pathways. This method ensures that trading decisions are driven by the underlying economics of feedstock costs and refining margins, enhancing the ability to systematically profit from inefficiencies in the pricing of petrochemical production alternatives.

4. Results and Parameter Stability

While many of the examined spread pairs exhibit linkages via their real optionality connections and historical cointegration, the practical trading outcomes show that most are ineffective for generating consistent returns. Despite their structural relationships, the behavior of these spreads

in real market conditions does not always translate into systematic profitability. A prevailing trend across both intra- and inter-spread strategies is that optimal configurations favor larger (ϵ) thresholds. This suggests that minor deviations in these spreads are rapidly corrected or efficiently priced by the market, leaving limited opportunities for profitable trades. Only larger deviations, which occur less frequently, appear to offer reliable trading opportunities. Furthermore, most of the profits from optimized parameter strategies were concentrated during periods of significant market upheaval—particularly during the COVID-19 pandemic's volatility shock. Outside of these periods, strategies using lower thresholds often stagnated or incurred losses, highlighting the importance of regime shifts in driving profitability.

The table below summarizes the optimal parameters for each intra-spread strategy, highlighting the configurations that achieved the highest Sharpe ratios.

Spread Pair	Optimized (n)	Optimized (ϵ)	Sharpe Ratio	Annualized P/L	RoD
Butane-Propylene	250	0.9	0.46	0.1497 \$/gal	2.9501
Propane-Propylene	250	0.75	0.47	0.1314 \$/gal	2.8064
Butane-Ethylene	150	0.6	0.56	0.1545 \$/gal	4.4627

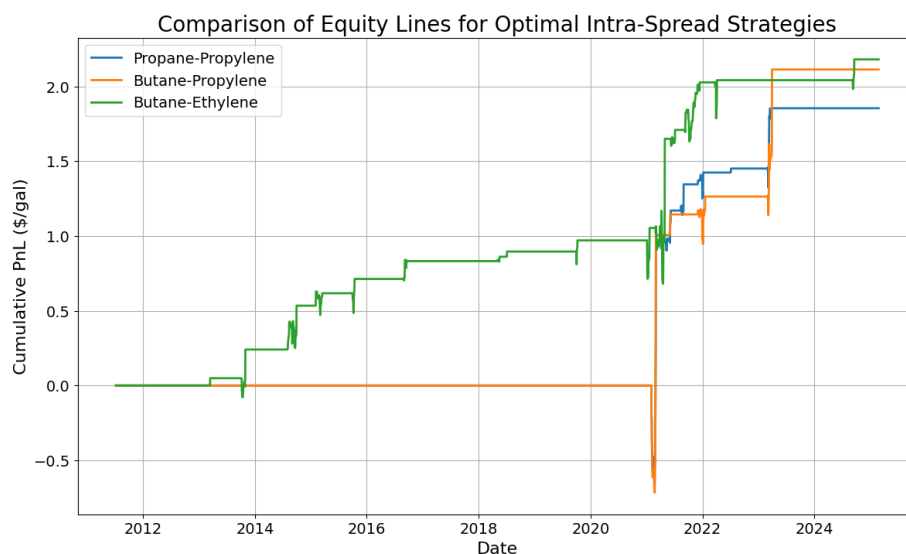


Figure 3: Intra-Spread Strategies

The equity curves for the intra-spread strategies reveal a clear divergence in performance between the tested pairs. Butane-Propylene, which remained largely inactive for extended periods, experienced a sharp drawdown before recovering in later years. Propane-Propylene, while more consistent, saw profitability concentrated in a few distinct periods rather than generating steady returns. The limited trading activity in these spreads indicates that most mispricings were short-lived and difficult to systematically capture.

The Butane-Ethylene spread on the other hand, presents a more dynamic trajectory. While it also exhibited periods of stagnation, it engaged in trades more frequently and showed more sustained equity growth, particularly in the years leading up to and immediately after the COVID-19 volatility shock. The sharp price dislocations during this period allowed the strategy to capitalize on extreme deviations before spreads reverted.

A key observation from the intra-spread results is that nearly all strategies exhibited minimal trading activity during the shale revolution period, where increased domestic production and refining efficiency compressed spreads across the market. This regime shift fundamentally altered the behavior of these pairs, reducing the likelihood of profitable divergences. The optimized thresholds (ϵ) reflect this reality, as higher values were necessary to trigger trades, effectively filtering out all but the most extreme mispricings. The rare occasions where these strategies engaged—such as during the COVID-19 volatility shock—suggest that they function more as tail-risk trades than systematic arbitrage opportunities.

Despite these broad limitations, the Butane-Ethylene spread stands out as a comparatively stronger candidate. Unlike the other two, its lower optimized threshold ($\epsilon = 0.6$) indicates a greater sensitivity to moderate deviations, allowing for more frequent engagement. The equity curve further supports this, displaying a steadier accumulation of returns rather than the sharp, irregular jumps seen in Butane-Propylene and Propane-Propylene. This suggests that while most intra-spreads fail to offer systematic profitability, select pairs like Butane-Ethylene may still hold some trading potential, particularly in periods of higher market volatility

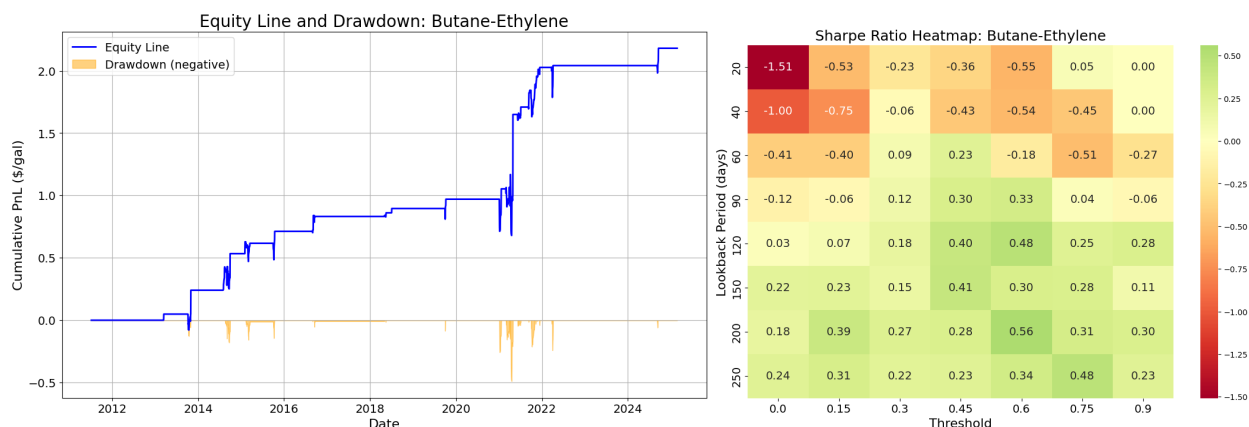


Figure 4: Butane-Ethylene Results

The back testing results for the inter-spread strategies further illustrate the complexities in identifying statistically sound yet practically viable trading pairs. Similar to the intra-spread results, statistical cointegration does not necessarily translate into consistently profitable trades. The following table summarizes the optimal parameters for each inter-spread strategy.

Spread Pair	(n)	(ϵ)	Sharpe Ratio	Annualized P/L	RoD
Ethane-Ethylene vs Propane-Propylene	250	0	0.64	0.6440 \$/gal	5.1889
Propane-Ethylene vs Propane-Propylene	200	0.64	0.55	0.3506 \$/gal	5.1293
Propane-Ethylene vs Butane-Ethylene	250	0.08	0.68	0.1157 \$/gal	6.5566
Propane-Propylene vs Butane-Ethylene	200	0.64	0.52	0.3353 \$/gal	4.4973
Butane-Ethylene vs Butane-Propylene	200	0.64	0.55	0.3506 \$/gal	5.1293

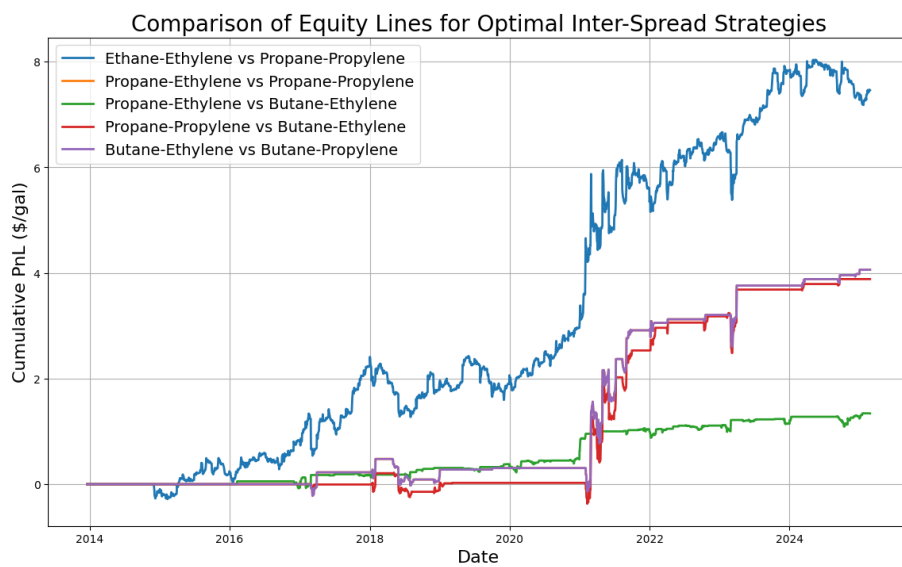


Figure 5: Inter-Spread Strategies

The inter-spread strategies offer a more compelling trading framework than their intra-spread counterparts, with several pairs demonstrating stronger Sharpe ratios and more consistent performance. Unlike intra-spreads, which relied on extreme threshold optimizations to trigger trades, inter-spreads remain profitable even at moderate thresholds, allowing for more systematic and frequent opportunities.

This profitability is driven by production pathway optionality in refining and petrochemical markets. While intra-spreads capture short-term mispricings within a fixed production process, inter-spreads reflect the broader ability of producers to shift between entirely different product pathways based on input costs and refining margins. This flexibility ensures that relative price spreads remain bounded over time, as market participants adjust production to capitalize on temporary distortions. As a result, inter-spread pairs exhibit more persistent mean reversion, making them a stronger foundation for systematic trading strategies.

The profitability of inter-spreads is further reinforced by their longer optimal lookback periods ($n = 200$ -250 days) and moderate trading thresholds ($\epsilon = 0$ -0.64). These findings suggest that pricing inefficiencies in production pathway flexibility take longer to resolve than simple intra-product dislocations but occur more frequently than the extreme mispricings driving intra-spread

strategies. This aligns with real-world observations that shifts in petrochemical production are dictated by gradual supply chain adjustments rather than short-term arbitrage.

The Ethane-Ethylene vs Propane-Propylene spread outperforms all others, with the highest Sharpe ratio and most consistent equity growth, reflecting its strong fundamental linkage. Unlike other spreads that require high thresholds to filter noise, this strategy performs best with no threshold, indicating that mispricings occur frequently and are systematically exploitable. This aligns with the fact that ethane and propane are the primary steam cracker feedstocks, while ethylene and propylene are their respective outputs, creating a direct arbitrage relationship driven by refining economics. Market participants constantly optimize between these inputs based on cost efficiency, ensuring persistent deviations that revert quickly. Additionally, the high liquidity and transparent pricing of these products make arbitrage more efficient, reinforcing the spread's tradability and long-term viability as a structured strategy.

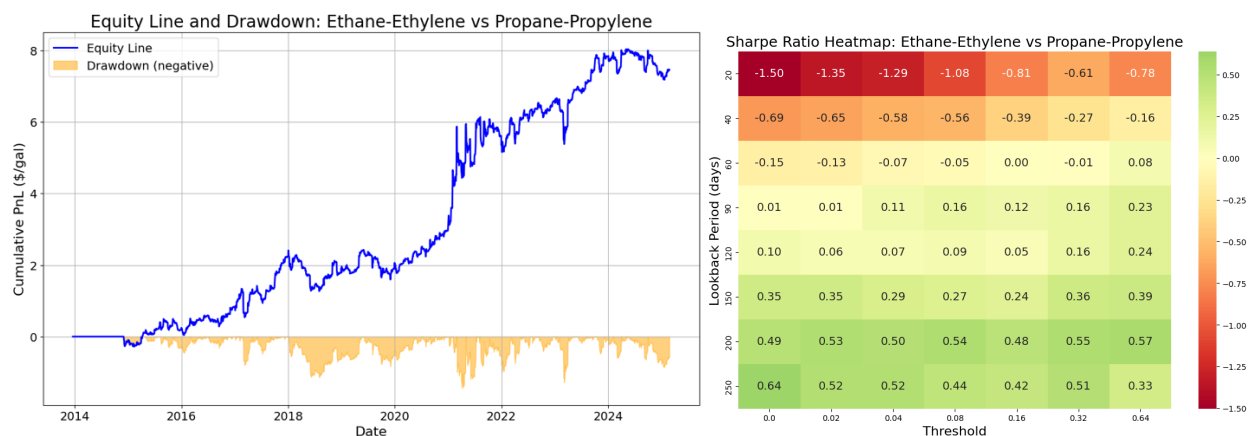


Figure 6: Ethane-Ethylene vs Propane-Propylene Strategy

5. Conclusion

While many spreads exhibit theoretical linkages through optionality and historical cointegration, only a few provide consistent trading opportunities. Intra-spread strategies tend to be highly selective, engaging only during extreme mispricings, while inter-spread strategies operate more systematically, capturing persistent pricing inefficiencies driven by production economics. The strongest strategies—Butane-Ethylene for intra-spreads and Ethane-Ethylene vs Propane-Propylene for inter-spreads—demonstrate frequent mean-reverting behavior and sustained profitability, making them the most viable candidates for structured trading. Given their outperformance, portfolio diversification across weaker spreads is ineffective, as it would dilute returns rather than enhance stability.

These findings suggest that while real optionality and cointegration are strong indicators of viable trading opportunities, long-term success hinges on the ability to adapt strategies to shifting market regimes and structural changes in production dynamics that influence spread behavior.